MARK 7980 Marketing Analytics & Decision Making
Spring 2025
Prof. Charlotte Mason

INSTRUCTOR

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CAPA Course Description

An analytical case-based course that emphasizes how to synthesize, analyze, and interpret data to support and guide marketing decisions in areas including promotion, advertising spending and copy, new product development, pricing, and loyalty programs.

Extended Course Description

Marketing is evolving from an art to a science. Recent years have witnessed increasing competitive pressures along with an explosion in the quantity and quality of marketing data available. Best-selling business books are promoting "Competing on Analytics", "Fact-based Marketing", "Data-driven Business Models", "The Numerati," and "Super-Crunchers – Why Thinking-by-Numbers is the New Way to Be Smart." A rising chorus is the scarcity of individuals with both business acumen as well as analytical instincts and training.

Whether a firm is consumer, business or service-oriented, acquiring and using information is critical for business planning and decisions. This analytical case-based course will emphasize how to analyze data to support and guide marketing decisions.

Many firms have extensive information, but far fewer have the expertise to act intelligently on such information. Data must be synthesized, analyzed and interpreted before sound marketing strategies and tactical plans can be developed. For example, marketers often need to understand/explain and predict the response to one or more marketing variables such as price, advertising, competitors' actions, or product changes. Beyond predicting response, it is vital to assess the financial impact of marketing decisions and programs.

The course uses a combination of lecture, cases, and exercises. The cases will be 'short on text' but hands-on in terms of analysis – meaning that, in order to answer the key case questions and decisions, you will be working with the 'raw' data rather than summary tables and exhibits. The course will emphasis interpretation and presentation of results. Examples of data and the accompanying decisions are:

- using customer transactional data from a casino to describe customer behavior & profitability
- data visualization using data from 84.51°'s extensive database of household supermarket purchases
- analyzing data from a test market to guide decisions for the national launch for a new consumer packaged good
- analyzing results of an A/B test in an online context
- analyzing historical data on advertising and sales to quantify the impact of advertising
- modeling customer lifetime value

- exploring different modeling techniques to predict response to a marketing offer
- using past customer transaction data to decide which customers to target for an online retailer's VIP program
- applying machine learning to data from a controlled market test to assess customer response to a credit card offer

We will use a combination of Excel (which I assume you are proficient with), IBM SPSS Statistics – a leading statistical analysis package which I assume will be new to most of you, and Tableau - a popular data visualization tool. You will need to download and install Tableau (free to students for 12-month license https://www.tableau.com/academic/students). In addition to SPSS, we may use JMP for some machine learning tools that are not available in the UGA licensed version of SPSS. JMP is a SAS software product that is free to UGA students to load on their computers, and can be obtained at https://eits.uga.edu/hardware_and_software/software/jmp/.

The course will be a combination of descriptive and predictive analytics. For descriptive analytics and data exploration, we will use IBM's SPSS statistical package and Tableau. For predictive analytics we will use both traditional statistical models such as regression as well as machine learning techniques. To assess the economic/profit implications, we will use Excel.

Required Prerequisite

MARK 7510 or permission of department

Goals

The principal objectives of the course are:

- To increase analytical skills through hands-on analysis of marketing data with commonly used marketing analytics techniques
- To link the analysis to key marketing decisions and profit impact
- To build your 'numerical fluency' and facility with common marketing metrics
- To improve your ability to think logically, analytically, and critically

Course Materials

- Required: MARK 7980 Coursepack from Bel-Jean
    - Purchase a digital only copy by going to this link: https://bel-jean.com/collections/spring-2025-academic-packets/products/mark-7980-mason-digital-only . You will be emailed a digital access code that is good for the duration of the semester
    - Important: you will need the coursepack because that is the only source for some of the copyrighted course materials and assignments
- Course Website: www.analytics.masonprof.com (also linked to eLC)
    - The link to the website is also posted in eLC along with the username and password
- I will make *limited* use of eLC – mostly for submitting assignments and I will also post datasets there in addition to the website. Please rely on the course website as the main source of information for the course.
- Tableau (free to students for 12-month license – you must use your uga.edu email address) https://www.tableau.com/academic/students
- IBM SPSS Statistics – there are several options
    - Access via the UGA virtual lab (this is free, but may not run as fast and is limited to 900 concurrent users) http://eits.uga.edu/support/vlab
    - Purchase a license (Windows or Mac) which can be downloaded on two computers (https://studentdiscounts.com/)
        - (recommended) 6-month license for SPSS GradPack Standard: $49 download

        - Other options:
            - 6-month license for SPSS GradPack Base: $35 download (this option does not include logistic regression which we will be using)
            - 12-month license for SPSS GradPack Premium: $89 download (more expensive but includes all (!) SPSS advanced modules)
    - Download a trial version good for 14 days: free https://www.ibm.com/products/spss-statistics
- (Recommended) SPSS Statistics Essential Training (free with your MyID)
    - Online course available at https://eits.uga.edu/learning_and_training/lynda/

To save on the cost of the coursepack, many required course readings are downloads from the course website. You may read these online, print a hard-copy, or save a copy. **Please note**: just because these are not in the coursepack does NOT mean they are optional reading!

PowerPoint slides for each class session will be posted on the course website. For many sessions, slides will be posted prior to class, otherwise slides will be posted after class.

You should rely on the course website as the primary source of information for the course and check it before each class for updates. Datasets for the cases are posted there – as well as additional (optional) readings and links. Major announcements will be posted on the homepage. If necessary, I will resort to mass email for any last minute or critical announcements.

Assignments and Grading Policy

The primary teaching philosophy of the course will be *learning by doing.* We will use a variety of tools to help understand the concepts and apply the tools and methods with direct hands-on applications. It is your responsibility to complete assigned readings before class and, when applicable, prepare any assigned questions. In many classes, we will work through examples and cases using SPSS or Tableau – so download the datasets before class and be prepared to follow along in class.

The course is structured in modules with an assignment for you to demonstrate your understanding and apply the material we have covered. Grades will be based on the following course assignments – all equally weighted:

Tableau Assignment (team)
SPSS Assignment (team)
A/B Testing Assignment (team)
Linear Regression Assignment (team)
Quiz #1 (individual)
Unsupervised Learning Assignment (team)
Logistic Regression Assignment (team)
Machine Learning/Decision Tree Assignment (team)
Quiz #2 (individual)
eTots case (individual)
Attendance and Participation

Assignments are due (submitted in eLC) on the assigned date. Points will be deducted for late assignments. You may complete assignments in self-selected teams of 2-3 members (or individually). You are free to switch up teams at any time. Any questions should be directed to me. Please note that I will try to answer specific questions – but I will not 'review' your assignments before they are submitted.

Please follow specific formatting instructions for the assignments. They vary a bit depending on the assignment but:

- Do not include the text of the assignment in your answers, Just submit your answers.
- Do not add a cover page – but do include your name(s) at the top of page 1.

- Do format into as few pages as possible while maintaining readability. This is because I will often print out your assignments and write comments directly on the printouts.

Quizzes: The quizzes will be a mixture of multiple choice, True/False, short (!) answer, and numerical problems. You should expect that they will take the full class period. I call them 'quizzes' rather than 'tests' because – while designed to cover all the material and many will find them challenging, they are designed not to be high stakes (meaning they count the same as each assignment). The quizzes will be closed book – but you may bring two sheets of notes (front & back, handwritten or a printout, whatever font you can read). My goal is for you to review the material, create your crib sheets, but not overly stress out. You will need a calculator.

To compute your overall numerical grade, I will rank everyone by their overall score and final grades will be based on that ranking. Note that there are no predetermined cutoffs for specific grades – rather, I will look for natural gaps to determine the cutoff for, say, an A versus an A-.

General Course Policies

- I expect everyone to attend class. You should arrive on time and not leave unless absolutely necessary; late entrants and coming/going during class are distracting.
- Use your name tents so that I can put faces and names together.
- Participation will be evaluated based on overall quality rather than quantity. You should do the reading, complete any assigned analysis, jot down responses to preparation questions. Try to make insightful comments and build on the discussion. Questions also 'count' as participation – if something is unclear to you, it likely is to other classmates as well. Be mindful of others and do not dominate the discussion.
- Arrangements to take the quizzes outside of the scheduled times must be approved in advance (minimum of 48 hours). If you are unexpectedly ill – you must notify me before the class.
- AI – unless explicitly permitted, the use of artificial intelligence software or word mixing software to write assignments or code, or disguise plagiarized work is considered unauthorized assistance
- For questions about specific grades, please email me with your specific questions or concerns.

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University Honor Code & Academic Honesty Policy

As a University of Georgia student, you have agreed to abide by the University's academic honesty policy*, "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others."* A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at www.uga.edu/ovpi. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Diversity and Inclusion

The Terry College is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

Changes to the Syllabus

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

Students with Disabilities

If a student needs academic assistance of any nature, the student must contact the Disability Resource Center. The professor cannot make accommodations for any student without DRC approval. Therefore, all

accommodations are managed by the Disability Resource Center. If you plan to request accommodations for a disability, please register with the Disability Resource Center. They can be reached by visiting Clark Howell Hall, calling 706-542-8719 (voice) or 706-542-8778 (TTY), or by visiting http://drc.uga.edu. If you plan to request accommodations, notify the instructor during the first two weeks of class.

Mental Health and Wellness Resources

- *UGA Well-being Resources:*

  UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

  Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

  UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

  - Well-being Resources: well-being.uga.edu
  - Student Care and Outreach: sco.uga.edu
  - University Health Center: healthcenter.uga.edu
  - Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
  - Health Promotion/ Fontaine Center: healthpromotion.uga.edu
  - Disability Resource Center and Testing Services: drc.uga.edu

  Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

Religious Accommodation Policy

University policy grants students excused absences from class or other organized activities or observance of religious holy days, unless the accommodation would create an undue hardship. You must notify me **by the end of the first week of classes** if you have any conflicts that may require an absence. It is your responsibility to make arrangements with me in advance to make up any missed work or in-class material and to coordinate with your team members to ensure your team is functioning properly

Tentative Course Schedule
Course Topics and Cases are listed below
Additional Details and Readings for Course Sessions are posted on the course website

1/6	**Course Introduction**
1/8	**Marketing Data, Regulations, Ethics and Privacy**
1/13	**Tableau I**
1/15	**Tableau II**
1/20	**MLK Holiday**
1/22	**Tableau III**
1/27	**SPSS I** **Due (team):** Tableau Assignment
1/29	**SPSS II**
2/3	**Business Experiments**
2/5	No Class – UGA Career Fair **Due (team):** SPSS Assignment
2/10	**Online A/B Advertising Testing** Case: Advertising Experiments at RestaurantGrades (coursepack)
2/12	**Analyzing Advertising Response: Time Series Analysis** Case: Alfonso's Department Store (coursepack)
2/17	**Applied Regression Review: Interactions, Non-linear Effects and Dummy Variables** **Due (team):** A/B Testing Assignment
2/19	**Multivariate Field Experiments: Ad Testing** Case: Nopane Advertising Strategy (coursepack)
2/24	**Multivariate Field Experiments: New Product Launch** **Due Before Class (team):** Newfood In-Class: Review of Regression assignment and review for Quiz
2/26	**Quiz #1**
	Spring Break
3/10	**Unsupervised Learning Techniques I** Factor Analysis/PCA
3/12	**Unsupervised Learning Techniques I** Cluster Analysis
3/17	**Customer Metrics I**
3/19	**Customer Metrics II: CLV** **Due (team):** Unsupervised Learning Assignment
3/24	**Customer Audit** Case: Madrigal: Conducting a Customer-Base Audit (coursepack)
3/26	**Recency, Frequency, Monetary (RFM) Segmentation**
3/31	**Predictive Modeling: Logistic Regression I**

4/2	**Predictive Modeling: Logistic Regression II** Case: Modern Farmhouse Designs: Targeting Customers with Logistic Regression
4/7	**Machine Learning: Decision Trees I** **Due (team):** Logistic Regression Assignment
4/9	**Machine Learning: Decision Trees II**
4/14	**Customer Loyalty Programs** Case (read only): Customer Care at eTots.com (coursepack) Identifying the VIPS at eTOTS (coursepack)
4/16	**Review and Q&A** **Due (team):** Machine Learning/Decision Tree Assignment
4/21	**Quiz #2**
4/23	**eTots workday**
4/27 (Sunday)	**Due (individual): etots.com decisions** Email me your decision file no later than 2 p.m.
4/28	**eTots Reveal & Parting Thoughts**